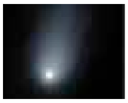
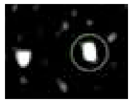
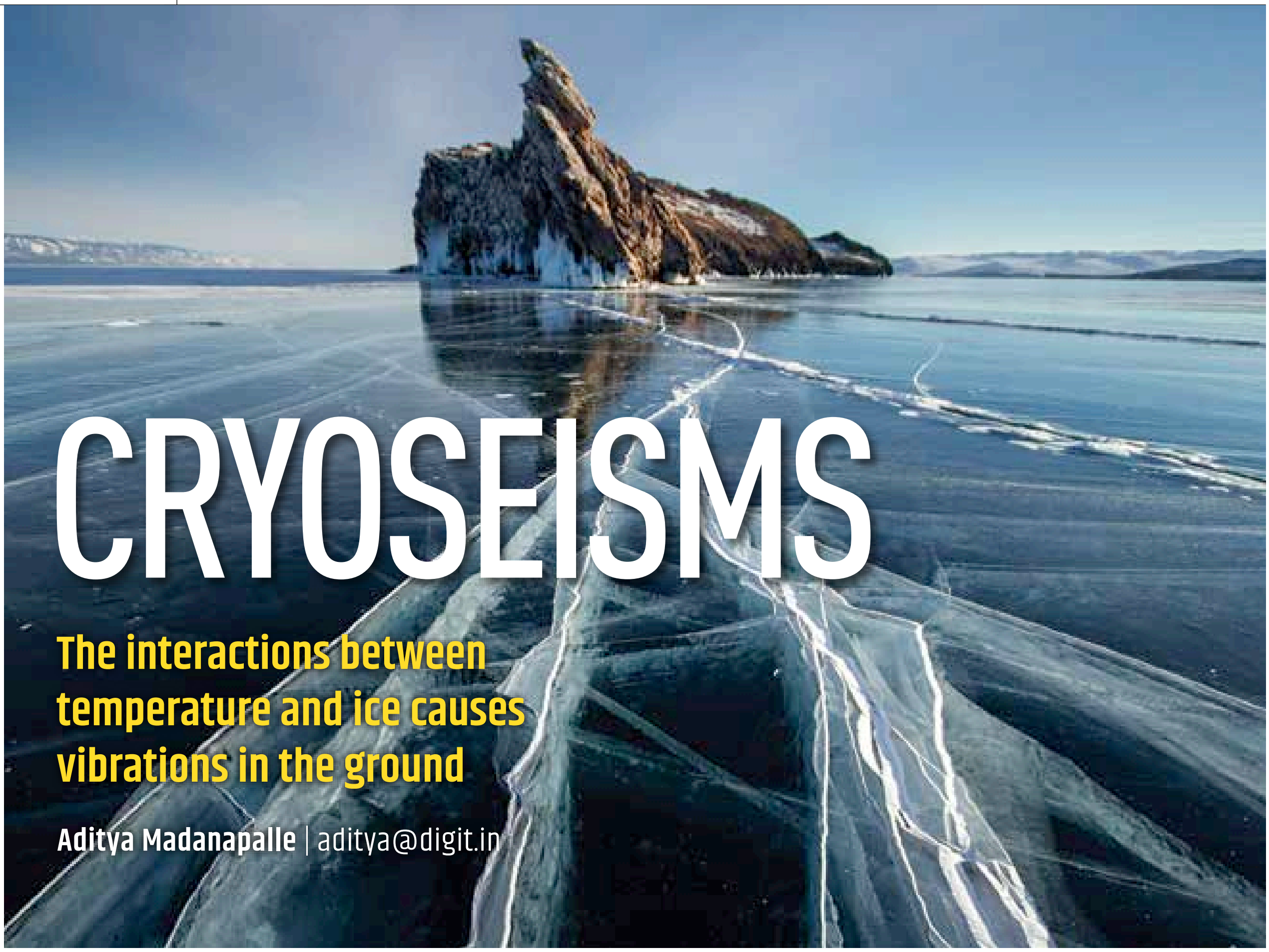


Interstellar Comet Borisov
Borisov, the second interstellar object known to humanity, is about to make its closest approach to Earth. **https://dgit.in/dec19-63**



A new type of Star System?
Mystery radio signals are hinting at a new type of star system in the galaxy: two stars orbiting each other. **https://dgit.in/dec19-64**



CRYOSEISMS

The interactions between temperature and ice causes vibrations in the ground

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Cryoseisms means the shaking of the earth caused by cold. There are two types of cryoseisms, frostquakes and icequakes. Although frostquakes and icequakes are used interchangeably at times, the two are very different phenomena, caused by very different geological processes. Considering how rare these phenomena are in nature, they are not well studied and are rarely recorded. Scientists are resorting to the use of status updates on social media platforms such as Twitter and Facebook to track, record and map frostquakes (but not icequakes). There is mounting evidence to suggest that cryoseisms of both types are increasing in frequency.

FROSTQUAKES

Frostquakes are a phenomena that is little understood. People who have experienced such an event note a sudden violent shaking event similar to an earthquake, but of a much shorter duration. Typically, frostquakes occur in the hours between midnight and dawn, when the temperatures are at their lowest. The quake may be accompanied by a loud boom or a thump. Small surface level cracks may appear, on ice, in snow deposits, or even in the ground. However, the disturbance is not picked up by seismometers, because they are not earthquakes at all. People living even a few meters away may not feel anything at all, as the quakes are extremely localised. While frostquakes can be scary, especially for the people who are jolted from their sleep in the middle of the night, they are not considered to be dangerous.

As such, frostquakes are not considered to be tectonic activity. They are caused when the temperature dips below the freezing point of water. If there is a lot of water below the ground, and the temperatures dip to freezing temperatures, then frostquakes can occur at such locations. Remember that water does not compress when it freezes, and instead expands after the state change from liquid to solid. The increase in pressure can be released violently, causing the loud booms, the ground shakes, and the cracks on the surface. These quakes can be felt on ice, and in areas where the ground has water ice mixed with rocks or soil. If the temperature falls well below the freezing point of water, from between -30 to -40 degrees Celsius, then another kind of frostquake can occur. Materials used for construction, such as cement and asphalt can contract, also leading to a build up of pressure. These materials can fracture in the intense cold. These events are also called frostquakes.

The earliest documented records of frostquakes may go back to 1819. Edward Hitchcock, an American Geolo-